

WASP-52b

R-1098

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_151105 · created 2026-07-30T18:52:43+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457331.7354 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.021
Transit depth (Rp/Rs)^2	3.02 +/- 0.73 [%]
Orbital Inclination (inc)	84.1 +/- 1.09
Airmass coefficient 1 (a1)	1.014 +/- 0.015
Airmass coefficient 2 (a2)	-0.0122 +/- 0.0096
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#3 - [623, 87]
Optimal Aperture	3.89
Optimal Annulus	3.89
Transit Duration (day)	0.066 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.174 $\pm$ 0.021 vs 0.1646 (deviation 0.45 σ)
Duration fit vs published	0.066 d vs 0.0758 d (deviation 0.89 σ)
Mid-time O-C	-5.8 min
Depth SNR	8.3
Residual scatter	1.47 %

Light curve

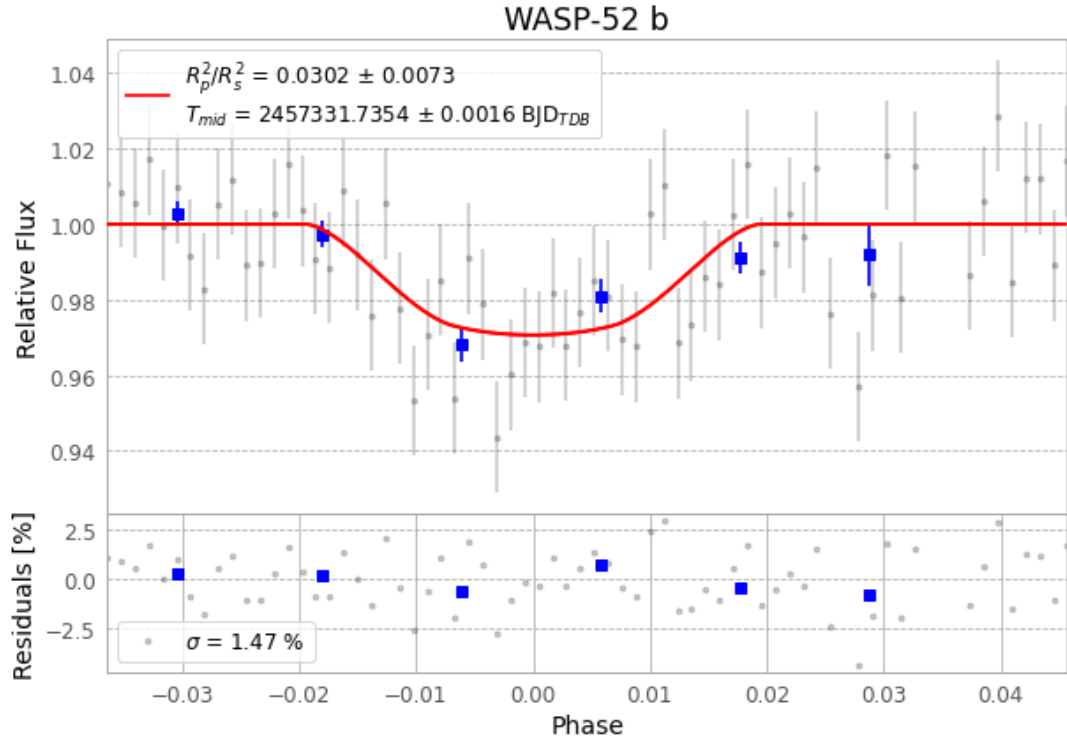


Figure 1 — FinalLightCurve_WASP-52 b_2015-11-04.png (SHA-256 123c6113635f4793...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [245, 242], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0c2a6dde6f780881...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-52151105040012.FITS → WASP-52151105072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-151105.log (0a81ee52a6c8...), FinalLightCurve_WASP-52 b_2015-11-04.png (123c6113635f...), FinalLightCurve_WASP-52 b_2015-11-04.csv (5f9ad303d17c...)

Compute resources

Wall time 310.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 18:52Z.